Evaluating Expressions with Radicals

Question Bank - Additional 4.05 **ANSWER KEY**

Name
Date
Period

Show your work. Circle your final answer.

For questions 1 through 6, evaluate.

1. $\sqrt{64}$

8

2. $\sqrt[3]{8}$

2

3. $\sqrt[3]{-125}$

−5

An odd root keeps the sign of its radicand.

4. $\sqrt{\frac{9}{49}}$

$\frac{3}{7}$

Top and bottom each get the root.

5. $\sqrt{4^2+33}$

7

16 + 33 = 49.

6. $\sqrt[3]{5^2+100}$

5

25 + 100 = 125.

For questions 7 and 8, evaluate. Each expression has a power and a radical.

7. Evaluate each expression.

a. $\sqrt{144} \cdot 3^{-2}$

$\frac{4}{3}$

$12 \cdot \frac{1}{9}$.

b. $\left(-\frac{1}{4}\right)^2 + \sqrt{3^2 + 27}$

$\frac{97}{16}$

$\frac{1}{16} + 6$.

c. $\sqrt[3]{-27} - 2^{-1}$

$-\frac{7}{2}$

$-3 - \frac{1}{2}$.

8. Evaluate each expression.

a. $\sqrt[3]{64} - 1.5\left(\sqrt{7^2 - 13}\right)$

−5

$4 - 1.5(6)$.

b. $\left(-\frac{1}{2}\right)^3 - \sqrt[3]{4^2 + 11}$

$-\frac{25}{8}$

$-\frac{1}{8} - 3$.

c. $\sqrt{6^2 + 13} \cdot 2^{-3}$

$\frac{7}{8}$

$7 \cdot \frac{1}{8}$.

Working backwards.

9. Find the value of n that makes each statement true. (The radicand is the unknown here, so it is not one of the listed perfect values until you find it.)

a. $\sqrt{n} = 15$

n = 225

b. $\sqrt[3]{n} = -2$

n = −8

c. $\sqrt{n-5} = 11$

n = 126

$n - 5 = 121$.

Reading someone else's work.

10. Deshawn writes $\sqrt{100-36} = \sqrt{100} - \sqrt{36} = 4$. Evaluate the expression correctly, and explain in one sentence what the radical bar does that his method ignores.

$\sqrt{100-36} = \sqrt{64} = 8$ ***. The bar groups the subtraction into a single radicand, so it is finished before any root is taken; the root of a difference is not the difference of the roots.***

His answer is half the true value. Ask him to try $\sqrt{25-9}$ *: his method gives 2 and the value is 4.*

11. Which is the value of $\sqrt{8^2+36}$?

- ○ A. 14
- ● **B. 10**
- ○ C. 100
- ○ D. 50

B

12. Select ALL of the expressions whose value is 4.

- ☐ A. $\sqrt{2^3+17}$
- ☒ **B.** $\sqrt[3]{64}$
- ☐ C. $\sqrt[3]{-64}$
- ☒ **D.** $\sqrt{16}$
- ☒ **E.** $\sqrt{5^2-9}$
- ☒ **F.** $\sqrt[3]{3^3+37}$

B, D, E and F